

Aman Anand

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Education

Indian Institute of Technology (Indian School of Mines) Dhanbad

Bachelor of Technology in Computer Science and Engineering (GPA: 8.20 / 10.00)

Expected May 2026

Dhanbad, Jharkhand

- **Relevant Coursework:** Artificial Intelligence, Data Structures and Algorithms (C++), OOPS(C++), Linear Algebra

Pace Junior Science college

PCM (Percentage: 87.83%)

May 2020 - April 2022

Mumbai, Maharashtra

Projects

NeRF Project | *Python, PyTorch, Computer Vision*

[Code]

- Replicated the Neural Radiance Fields (NeRF) model based on the original research paper for 3D scene reconstruction.
- Implemented a vanilla NeRF model using PyTorch, leveraging volume rendering techniques for novel view synthesis.
- Trained the model on the synthetic LEGO dataset and generated novel views for evaluation.
- Followed research insights and online resources to improve the replication process.

Autonomous Buggy (NXP AIM) | *NavQ+, Python, ROS, Gazebo, YOLOv5*

[Video]

- Simulated a buggy on Gazebo UI to autonomously navigate through its environment using a camera and LiDAR.
- Completed the line-following track with obstacles in the simulation within the 2-minute time limit.
- Successfully deployed the algorithm on the Hardware Buggy with added functionality of traffic sign detection.
- Implemented YOLOv5 model on the NavQ+ processor to segment and classify traffic signs and take actions accordingly.

Hologlyph Bots (E-Yantra) | *Python, ROS2, Gazebo, Linux, SolidWorks*

[Video]

- Designed the bot using SolidWorks and generated G-Code for 3D printing with Cura Ultimaker.
- Integrated inverse kinematics for planar translation of a 3-wheeled holonomic drive with continuous rotation servos
- Implemented Aruco Detection for pose extraction of mounted pens on the robot against the arena.
- Developed a PID control loop with camera feedback for precise motion control.
- Used Micro-ROS to convert an ESP-32 into a ROS2 node for communication and controlled the writing action using a servo motor.
- Prepared the Gazebo simulation of the bot along with an overhead camera integrated with the arena.

Technical Skills

Languages: Python, MATLAB, C++

Technologies: Pytorch, Langchain, HuggingFace, Kaggle, Google Colab, Linux, Simulink, ROS2, Solidworks

Concepts: Machine Learning, Computer Vision, Mechanics, Control Systems, Object-Oriented Programming, Data Structures and Algorithms .

Achievements

- Won the Regional Finale for NXP AIM competition
- Bagged 2nd Runner-Up position at TechKriti'24 RoboWar, IIT Kanpur.
- Ranked among the top 50 teams in the E-yantra Robotics Competition

Social Engagements

Club-member : at RoboISM - Robotronics club of IIT Dhanbad

Volunteer:

- At Kartavya - NGO run by students of IIT Dhanbad to educate underprivileged childrens, uplift the society.
- At Prakash'25 (Annual day of kartavya) hospitality team.

Hobbies: Book reading, Sketching